MTH301

Midterm Important Questions 2024

VU360

- ***Intersection of two straight lines***: *The intersection of two straight lines is a point.*
- ***Plane***: *A plane is a two − dimensional surface.*
- ***Chain rule***: *If* $w = f(x,y,z)$ *and* $x = g(r,s), y = h(r,s), z = t(r,s)$ *then by the chain rule,* $\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} * \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} * \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} * \frac{\partial z}{\partial r}$.
- ***Parametric equations***: *The parametric equations that*
- *correspond to the given situation are* $x = x(r,s), y = y(r,s), z = z(r,s)$.
- ***Derivative of a vector − valued function***: *The derivative of the vector − valued function* $r(t) = (\cos(5t), \tan(t), 6\sin(t))$ *is* $r'(t) = (-5\sin(5t), \sec\text{^}2(t), 6\cos(t))$.
- ***Exact differential***: *The differential* $dz = (x^2 * y + y)dx - x\,dy$ *is exact.*
- ***Symmetry***: *If the equation of a curve in polar*

  coordinates remains unchanged after replacing (r,θ) *by* $(r, \pi - \theta)$, *then the curve is symmetric about the* $y-$ *axis.*

- ***Amplitude***: *The amplitude of the periodic function* $f(x) = \sin(3x)$ *is* 1.
- ***Period***: *The period of the periodic function whose graph is given is* 4.
- ***Laplace Transform***: *The Laplace transform of the function* $F(t) = t$ *is* $\frac{1}{s^2}$.
- ***Line integral***: *The line integral* $\int (2x + y)dx + (x - y)dy$ *over the line segment from* $(0,0)$ *to* $(2,0)$ *is* 4.
- ***Direction ratios***: *The direction ratios for the line joining the points*

  $(1,3,5)$ *and* $(2,-1,4)$ *are* $1, -4,$ *and* -1.

- ***Gradient***: *The gradient of the function* $f(x,y) = x^{2y}$ *is* $(2xy, x^2)$.
- ***Double integral***: *The double integral* $\int\int (x^2 + y^2)dx\,dy$ *over the region bounded by the lines* $x = 0, y = 0,$ *and* $x + y = 1$ *is* $\frac{1}{3}$.
- ***Vector field***: *The vector field* $F(x,y) = (x,y)$ *is conservative.*
- ***Curl***: *The curl of the vector field* $F(x,y,z) = (z,x,y)$ *is* $(1,1,1)$.
- ***Divergence***: *The divergence of the vector field* $F(x,y,z) = (x,y,z)$ *is* 3.
- ***Line integral***: *The line integral* $\int (x^2 - y^2)dx + (2xy)dy$ *over the circle* $x^2 + y^2 = 4$ *is* 0.
- ***Green's theorem***: *The circulation of the vector field* $F(x,y) = (x,y)$ *around the unit circle is* 2π.
- ***Parametric equations***: *The parametric equations for the curve* $y = 2x^2 - 3x + 1$ *are* $x = t, y = 2t^2 - 3t + 1$.
- ***Polar coordinates***: *The polar coordinates for the point* $\left(3, \frac{\pi}{4}\right)$ *are* $\left(3, \frac{\pi}{4}\right)$.

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- ***Laplace transform***: *The Laplace transform of the function* $F(t) = t^2$ *is* $\frac{2}{s^3}$.
- ***Fourier series***: *The Fourier series of the function* $f(x) = x$ *is* $\left(\frac{\pi}{2}\right) - \left(\frac{4}{\pi}\right)\left(\frac{1}{n^2}\right)\cos\sum (nx)$.

- ***Vector calculus***: *Find the divergence and curl of the vector field* $F(x, y, z) = (x^2, y^2, z^2)$.
- ***Double integral***: *Evaluate the double integral* $\int\int (x^2 - y^2)dx\,dy$ *over the region bounded by the lines* $x = 0, y = 0,$ *and* $x + y = 2$.
- ***Line integral***: *Find the line integral* $\int (x + y)dx + (x - y)dy$ *over the curve* $C: x = t, y = t^2$ *from* $(0,0)$ *to* $(1,1)$.
- ***Green's theorem***: *Use Green's theorem to find the circulation of the vector field* $F(x, y) = (x^2, y^2)$ *around the unit circle.*
- ***Parametric equations***: *Find the parametric equations for the curve* $y = x^3 - 3x^2 + 2x + 1$.
- ***Polar coordinates***: *Convert the rectangular coordinates* $\left(2, \frac{\pi}{3}\right)$ *to polar coordinates.*
- ***Laplace transform***: *Find the Laplace transform of the function* $F(t) = t^3$.
- ***Fourier series***: *Find the Fourier series of the function* $f(x) = x^2$.
- ***Partial differential equations***: *Solve the partial differential equation* $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2x + 3y$.
- ***Gradient***: *Find the gradient of the function* $f(x, y) = x^{3y^2}$.
- ***Directional derivative***: *Find the directional derivative of the function* $f(x, y) = x^{2y}$ *at the point* $(1,2)$ *in the direction of the vector* $(1,1)$.

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